

Model Governance & Validation Framework

Auto-Modeller Bayesian Marketing Mix Model (MMM)

1. Purpose of Model Governance

Model governance ensures that the MMM system is:

- **reliable**
- **auditable**
- **reproducible**
- **consistent across clients**
- **aligned with business decision-making**

The governance framework defines how models are:

- validated
 - approved
 - monitored
 - retrained
 - audited
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2. Governance Principles

The MMM platform follows five governance principles:

1. Transparency

All modelling assumptions and transformations are documented.

2. Reproducibility

Every model run must be reproducible using the same dataset and configuration.

3. Data Integrity

Input data undergoes strict validation before modelling.

4. Model Stability

Models must pass diagnostic checks before being accepted.

5. Auditability

Model outputs and configuration are archived for review.

3. Model Lifecycle

The MMM model lifecycle includes the following stages:

Data Collection



Data Validation



Model Training



Diagnostics



Business Validation



Deployment



Monitoring



Retraining

4. Data Governance

4.1 Input Data Sources

Typical sources include:

Source	Description
CRM systems	marketing exposures
Media platforms	campaign activity
Sales systems	revenue or volume
External data	economic indicators

4.2 Data Validation Rules

Before modelling begins, the following checks are executed.

Schema Validation

Ensures required columns exist.

Required columns:

date

region

sub_brand

variable

value

Variable Consistency

Ensures variables listed in **info.csv** match those in **data.csv**.

Missing Value Checks

The system flags:

- missing KPI values
 - missing media exposures
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Variation Checks

Marketing variables must exhibit **sufficient variation**.

Channels with no variation cannot be modeled effectively.

Time Continuity Check

Ensures no gaps exist in the time series.

5. Model Training Governance

5.1 Model Frequency Standardization

All models are trained at **weekly frequency**.

If input data is monthly:

monthly → weekly conversion

is performed while preserving totals.

5.2 Automatic Stability Controller

The Auto-Modeller uses **stability levels** to prevent estimation failures.

Level Description

Level 0 Full flexibility

Level 1 Moderate constraints

Level 2 Strong constraints

If instability occurs:

model automatically retries with tighter priors

5.3 Bayesian Sampling Governance

Sampling configuration:

chains = 4

draws = 1000

tune = 1000

target_accept = 0.95

This configuration balances:

- estimation accuracy
 - computational efficiency
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6. Model Diagnostic Requirements

A model must pass diagnostic tests before approval.

6.1 Divergence Check

Hamiltonian Monte Carlo divergences indicate sampling problems.

Acceptable threshold:

divergences < 100

If exceeded:

- priors are tightened
 - model retraining is triggered
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6.2 Effective Sample Size

Effective sample size must exceed:

$$ESS > 100$$

for key parameters.

6.3 Convergence Check

Convergence metric:

$$R\text{-hat} < 1.05$$

If violated, model retraining is required.

6.4 Predictive Accuracy

Key performance metrics:

Metric Purpose

RMSE prediction error

MAPE relative error

R^2 variance explained

Typical acceptable ranges:

Metric Expected Range

MAPE < 30%

R^2 > 0.3

These thresholds may vary by industry.

7. Economic Sanity Checks

To prevent unrealistic results, economic rules are applied.

7.1 ROI Bounds

Channels with extremely high ROI are flagged.

Example threshold:

ROI > 50 → warning

7.2 Contribution Consistency

Marketing contributions must not exceed total KPI.

If:

Σ media contribution > KPI

reconciliation adjustment is applied.

7.3 Spend-Contribution Relationship

Channels with negligible spend but extremely high contribution are flagged for review.

8. ROI Stability Diagnostics

ROI uncertainty is evaluated using:

$$StabilityRatio = \frac{ROI_{p95} - ROI_{p05}}{|ROI_{mean}|}$$

Classification:

Stability Ratio Interpretation

< 1 stable

1–2 moderate

2 | unstable |

Channels marked **unstable** should be interpreted cautiously.

9. Model Approval Process

Before results are shared with business stakeholders, the following steps occur:

Step 1

Technical diagnostics review.

Step 2

Economic sanity checks.

Step 3

Contribution plausibility assessment.

Step 4

Business validation with domain experts.

10. Model Output Governance

All model runs generate standardized outputs.

Primary outputs include:

roi.csv

media_contribution.csv

base_contribution.csv

marginal_roi_curves.csv

roi_key_points.csv

predictions_timeseries.csv

fit_metrics.csv

These outputs are archived for audit purposes.

11. Model Versioning

Each model run is associated with:

Attribute	Description
dataset_version	data snapshot
model_version	code version
run_timestamp	execution time
configuration	modelling parameters

This enables full reproducibility.

12. Model Monitoring

Once deployed, models should be monitored for:

- KPI drift
- marketing mix changes
- structural market changes

Indicators of model degradation include:

forecast errors increasing
unexpected ROI shifts
new marketing channels

13. Model Retraining Policy

Recommended retraining frequency:

Scenario	Retraining Frequency
stable marketing strategy	quarterly
rapidly changing markets	monthly
major campaign launches	immediate retraining

14. Data Retention Policy

Model inputs and outputs should be retained for:

minimum 2 years

This supports audit requirements and longitudinal analysis.

15. Security and Access Control

Access to modelling systems should follow role-based permissions.

Role	Permissions
admin	full system access
modeller	run models
analyst	view results
business user	dashboards only

16. Audit Trail

Each model run should log:

input files
model configuration
sampling diagnostics
generated outputs

These logs ensure complete traceability.

17. Limitations and Responsible Use

Users should be aware of limitations:

- MMM is based on historical data
 - results depend on data quality
 - causal interpretation requires sufficient variation
 - results should complement, not replace, business judgment
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18. Governance Responsibilities

Role	Responsibility
Data Engineering	data integrity
Modelling Team	model development
Analytics Lead	model approval
Business Stakeholders	interpretation